



Question Bank

Course Title: Discrete Mathematical Structures and Graph Theory (BCSESC404A)

Module 1: Fundamentals of Logic

Q. No.	Written Type Question	Marks	RL	CO
1	Define the following with an example for each: i) Disjunction ii) Conjunction iii) Negation	6	L1	1
2	Define the following with an example for each: i) Proposition ii) Tautology iii) Contradiction	6	L1	1
3	Let p, q and r be propositions having truth values 0, 0 and 1 respectively. Identify the truth values of the following compound propositions: i) $(p \wedge q) \rightarrow r$ ii) $p \rightarrow (q \wedge r)$ iii) $p \wedge (r \rightarrow q)$ iv) $p \rightarrow (q \rightarrow (\neg r))$.	6	L3	2
4	Verify the compound proposition for tautology and contradiction $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$	6	L2	2
5	Verify the compound proposition for tautology and contradiction $[(p \vee q) \wedge \{(p \rightarrow r) \wedge (q \rightarrow r)\}] \rightarrow r$	6	L2	2
6	Verify the compound proposition for tautology and contradiction $(p \vee q) \rightarrow r \leftrightarrow (\neg r \rightarrow \neg (p \vee q))$	6	L2	2
7	Verify the compound proposition for tautology and contradiction $\{p \rightarrow (q \rightarrow r)\} \rightarrow \{(p \rightarrow q) \rightarrow (p \rightarrow r)\}$	6	L2	2
8	Verify $(p \vee q) \leftrightarrow (p \vee q) \wedge \neg (p \wedge q)$ using truth tables.	6	L2	2
9	Verify $[(p \rightarrow (q \rightarrow r))] \leftrightarrow [(p \wedge \neg r) \rightarrow \neg q]$ using truth tables.	7	L2	2
10	Verify the compound propositions $p \wedge ((\neg q) \vee r)$ and $p \vee (q \wedge (\neg r))$ are logically equivalent or not.	7	L2	2
11	Solve the compound proposition $p \vee q \wedge [\neg \{(\neg p) \wedge q\}]$ using laws of logic.	7	L2	2
12	Solve the compound proposition $p \vee q \wedge [\neg \{(\neg p) \vee q\}]$ using laws of logic.	7	L2	2
13	Prove the logical equivalence $[p \vee q \vee (\neg p \wedge \neg q \wedge r)] \leftrightarrow p \vee q \vee r$ using laws of logic.	7	L2	2
14	Prove the logical equivalence $p \rightarrow (q \rightarrow r) \leftrightarrow (p \wedge q) \rightarrow r$ using laws of logic.	7	L2	2
15	State the converse, inverse and contrapositive of: i) If the triangle is not isosceles, then it is not equilateral ii) If the real number x^2 is greater than zero, then x is not equal to zero.	7	L1	1

Q. No.	Written Type Question	Marks	RL	CO
	iii) If a quadrilateral is a parallelogram, then its diagonals bisect each other.			
16	Test whether the following is valid argument. If Sachin hits a century, then he gets a free car. <u>Sachin hits a century.</u> $\therefore$ Sachin gets a free car.	7	L4	2
17	Test the validity of the following arguments. If Ravi goes out with friends, he will not study. If Ravi does not study, his father will become angry. <u>His father is not angry.</u> $\therefore$ Ravi has not gone out with friends.	7	L4	2
18	Test the validity of the following argument I will get grade A in this course or I will not graduate. If I do not graduate, I will join army. <u>I got grade A.</u> $\therefore$ I will not join army.	7	L4	2
19	Test whether the following arguments are valid $p \rightarrow q$ $r \rightarrow s$ <u>$p \vee r$</u> $\therefore q \vee s$	6	L4	2
20	Test whether the following arguments are valid $p \rightarrow q$ $r \rightarrow s$ <u>$\neg q \vee \neg s$</u> $\therefore \neg (p \wedge r)$	6	L4	2
21	Define Open statement, Quantifiers and types of Quantifiers.	6	L1	1
22	Negate and simplify each of the following. (i) $\exists x, [p(x) \vee q(x)]$ (ii) $\forall x, [p(x) \wedge \neg q(x)]$ (iii) $\forall x, [p(x) \rightarrow q(x)]$ (iv) $\exists x, [p(x) \vee q(x)] \rightarrow r(x)$	7	L2	2
23	Let $p(x): x^2 - 7x + 10$, $q(x): x^2 - 2x - 3$, $r(x): x < 0$. Determine the truth or falsity of the following statements. When the universe U contains only the integers 2 and 5. If a statement is false. Provide a counter example or explanation. (i) $\forall x, [p(x) \rightarrow \neg r(x)]$ (ii) $\forall x, [q(x) \rightarrow r(x)]$ (iii) $\exists x, [q(x) \rightarrow r(x)]$ (iv) $\exists x, [p(x) \rightarrow r(x)]$	7	L3	2
24	Write down the following proposition in symbolic form and determine its negation: “If all triangles are right angled, then no triangle is equiangular”.	7	L3	2

Q. No.	Written Type Question	Marks	RL	CO
25	Verify whether the following argument is valid. If a triangle has 2 equal sides, then it is isosceles. If the triangle is isosceles, then it has 2 equal angles. <u>A certain triangle ABC does not have 2 equal angles.</u> $\therefore$ the triangle ABC does not have 2 equal sides.	7	L2	2
26	Prove that the following argument is valid. $\forall x, [p(x) \vee q(x)]$ $\exists x, \neg p(x)$ $\forall x, [\neg q(x) \vee r(x)]$ <u>$\forall x, [s(x) \rightarrow \neg r(x)]$</u> $\therefore \exists x, \neg s(x)$	7	L2	2
27	Prove that, for all integers k and l, if k and l are both odd then $k + l$ is even and kl is odd.	7	L2	2
28	For each of the following statements, provide an indirect proof by stating and proving the contrapositive of the given statement. (i) for all integers k and l, if kl is odd then both k and l are odd. (ii) for all integers k and l if $k + l$ is even, then k and l are both even or both odd.	7	L2	2
29	Give (i) direct proof (ii) indirect proof (iii) proof by contradiction for the following statement: “if n is an odd integer, then $n + 9$ is an even integer”.	7	L2	2
30	Prove that every even integer n with $2 \leq n \leq 26$ can be written as a sum of most three perfect squares.	7	L2	2

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Course Title: Discrete Mathematical Structures and Graph Theory (BCSESC404A)

Module 2: Properties of Integers

Q. No.	Written Type Question	Marks	RL	CO
1	Make use of mathematical induction and prove that, for all positive integers $n \geq 1$, $1 + 2 + 3 + \dots + n = \frac{1}{2}n(n+1)$	6	L2	2
2	Make use of mathematical induction and prove that, for each $n \in \mathbb{Z}^+$, $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{[n(n+1)(2n+1)]}{6}$	6	L2	2
3	Make use of mathematical induction and prove that every positive integer $n \geq 24$ can be written as a sum of 5's and/or 7's.	6	L2	2
4	Make use of mathematical induction and prove that, for all integers $n \geq 1$ $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{[n(2n-1)(2n+1)]}{3}$	6	L2	2
5	Make use of mathematical induction and prove that, for all positive integers $n \geq 1$, $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + n(n+1) = \frac{1}{3}n(n+1)(n+2)$	6	L2	2
6	Make use of mathematical induction and prove that, for every positive integer n , 5 divides $n^5 - n$	6	L2	2
7	A bit is either 0 or 1. A byte is a sequence of 8 bits. Determine (i) the number of bytes (ii) the number of bytes that begin with 11 and end with 11 (iii) The number of bytes that begin with 11 and do not end with 11 (iv) the number of bytes that begin with 11 or end with 11.	7	L3	2
8	License plates consist of 2 letters followed by 4 digits. Determine in how many ways can this be done if (i) No letter or digit can be repeated (ii) Repetitions are allowed (iii) Only vowels and even digits can be used (with repetitions)	7	L3	2
9	Determine the number of proper divisors of 441000.	6	L3	2
10	Determine the number of permutations of the letters of the word MASSASAUGA. In how many of these, all four 'A's are together? How many of them begin with S?	7	L3	2
11	Determine how many arrangements are there for all letters in the word SOCIOLOGICAL and how many of these arrangements A and G are adjacent?	7	L3	2
12	Determine how many Positive integers n can we form using the digits 3, 4, 4, 5, 5, 6, 7 if we want n to exceed 5,000,000	7	L3	2
13	A woman has 11 close relatives and she wishes to invite 5 of them to dinner. Determine in how many ways can she invite them in the following situations: (i). There is no restriction on the choice. (ii). Two particular persons will not attend separately. (iii). Two particular persons will not attend together	7	L3	2

Q. No.	Written Type Question	Marks	RL	CO
14	Determine the number of arrangements of all the letters in TALLAHASSEE. How many of these arrangements have no adjacent A's?	7	L3	2
15	A committee of 12 is to be selected from 10 men and 10 women. Determine in how many ways can the selection be carried out if (a) there are no restrictions (b) there must be six men and six women (c) there must be an even number of women	7	L3	2
16	A man has 15 close friends of whom 6 are women. Determine in how many ways can he invite 3 or more of these friends to a party if he wants the same number of men(including himself) as women.	7	L3	2
17	Determine how many bytes contain (i) Exactly two 1's (ii) Exactly four 1's (iii) Atleast six 1's.	6	L3	2
18	Determine the coefficient of (i) x^9y^3 in the expansion of $(2x - 3y)^{12}$ (ii) x^{12} in the expansion of $x^3(1 - 2x)^{10}$	6	L3	2
19	Determine the coefficient of (i) xyz^2 in the expansion of $(2x - y - z)^4$ (ii) $x^{11}y^4$ in the expansion of $(2x^3 - 3xy^2 + z^2)^6$	6	L3	2
20	Determine the coefficient of (i) $a^2b^3c^2d^5$ in the expansion of $(a + 2b - 3c + 2d + 5)^{16}$ (ii) $w^3x^2yz^2$ in the expansion of $(2w - x + 3y - 2z)^8$	6	L3	2
21	Determine in how many ways can we distribute 7 apples and 6 oranges among 4 children so that each child gets atleast 1 apple?	7	L3	2
22	Determine in how many ways can one distribute eight identical balls into four distinct containers so that (i) no container is left empty (ii) the fourth container gets an odd number of balls	7	L3	2
23	Determine the number of positive integer solutions of the equation $x_1 + x_2 + x_3 = 17$	6	L3	2
24	Determine the number of positive integer solutions of the equation $x_1 + x_2 + x_3 + x_4 + x_5 = 30$ where $x_1 \geq 2, x_2 \geq 3, x_3 \geq 4, x_4 \geq 2, x_5 \geq 0$	6	L3	2
25	Determine in how many ways can 10 identical pencils be distributed among 5 children in the following cases: i) There are no restrictions. ii) Each child gets atleast one pencil. iii) The youngest child gets atleast two pencils.	7	L3	2

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Module 3: The Principle of Inclusion and Exclusion

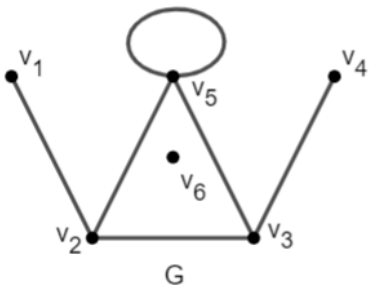
Q. No.	Written Type Question	Marks	RL	CO									
1	Out of 30 students in a hostel, 15 study History, 8 study Economics and 6 study Geography. It is known that 3 students study all these subjects. Show that 7 or more students study none of these subjects.	6	L2	3									
2	Determine in how many ways 5 number of a's, 4 number of b's and 3 number of c's can be arranged so that all the identical letters are not in a single block .	6	L3	3									
3	Determine how many integers between 1 and 300(inclusive) are (i) divisible by atleast one of 5,6,8 (ii) divisible by none of 5,6,8	6	L3	3									
4	Determine in how many ways can 26 letters of the English alphabets be permuted so that none of the patterns CAR, DOG, PUN (or) BYTE occurs.	7	L3	3									
5	Determine in how many ways can one arrange the letters in the word CORRESPONDENTS so that (i) There is no pair of consecutive identical letters (ii) There are exactly two pairs of consecutive identical letters (iii) There are atleast three pairs of consecutive identical letters	7	L3	3									
6	Determine in how many ways can each of 10 people select a left glove and a right glove out of total 10 pairs of gloves so that no person selects a matching pair of gloves.	6	L3	3									
7	While at the race track a person bet on each of the 10 horses in a race to come in accordance to how they are favoured. Determine in how many ways can they reach the finish line so that he loses all his bets.	7	L3	3									
8	There are 8 letters to 8 different people to be placed in 8 different address envelopes. Determine the number of ways of doing this so that atleast one letter gets to the right person.	7	L3	3									
9	For the positive integers 1,2,3,,n there are 11660 derangements where 1,2,3,4,5 appear in the fist five positions. Determine the value of n.	6	L3	3									
10	Determine the rook polynomial for the board shown below <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td>1</td><td>2</td><td></td></tr><tr><td></td><td></td><td>3</td></tr><tr><td>4</td><td>5</td><td>6</td></tr></table>	1	2				3	4	5	6	7	L3	3
1	2												
		3											
4	5	6											
11	Determine the rook polynomial for the board shown below <table border="1" style="margin-left: auto; margin-right: auto;"><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td></td><td>5</td></tr><tr><td>6</td><td>7</td><td>8</td></tr></table>	1	2	3	4		5	6	7	8	7	L3	3
1	2	3											
4		5											
6	7	8											

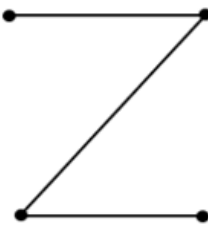
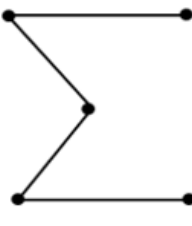
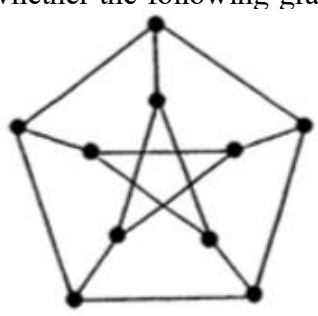
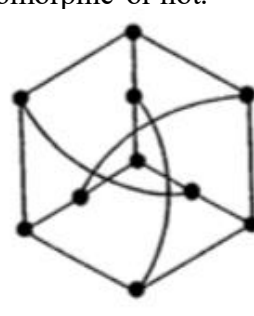
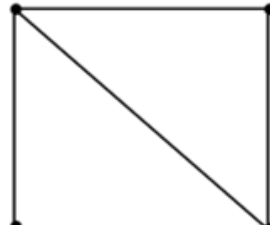
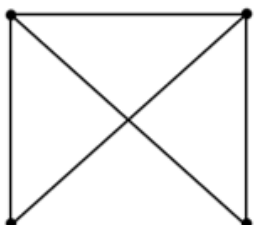
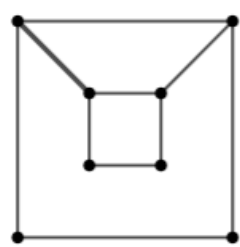
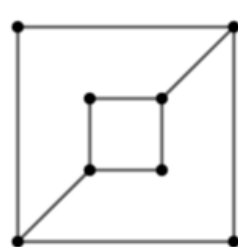
Q. No.	Written Type Question	Marks	RL	CO												
12	Make use of expansion formula and determine the rook polynomial for the 3 * 3 board.	7	L3	3												
13	Make use of expansion formula and obtain the rook polynomial for the board shown below <table><tr><td></td><td></td><td>1</td></tr><tr><td></td><td>2</td><td>3</td></tr><tr><td>4</td><td>5</td><td>6</td></tr><tr><td>7</td><td>8</td><td></td></tr></table>			1		2	3	4	5	6	7	8		7	L3	3
		1														
	2	3														
4	5	6														
7	8															
14	An apple, a banana, a mango and an orange are to be distributed to four boys B ₁ , B ₂ , B ₃ , B ₄ . The boys B ₁ and B ₂ do not wish to have apple, the boy B ₃ does not want banana or mango and B ₄ refuse orange. Determine in how many ways the distribution can be made so that no boy is displeased.	7	L3	3												
15	Five teachers T ₁ , T ₂ , T ₃ , T ₄ , T ₅ are to be made class teachers for five classes, C ₁ , C ₂ , C ₃ , C ₄ , C ₅ one teacher per each class. T ₁ and T ₂ do not wish to become the class teacher for C ₁ or C ₂ , T ₃ and T ₄ for C ₄ or C ₅ and ,T ₅ for C ₃ or C ₄ or C ₅ . Determine in how many ways can the teachers be assigned the work. (Without displacing any teacher)	7	L3	3												
16	Determine the recursive definition for the sequence {a _n } is each of the following cases. (i)a _n = 5n (ii)a _n = 6 ⁿ (iii)a _n = n(n + 2) (iv) a _n = n ²	7	L3	3												
17	For the Fibonacci sequence F ₀ , F ₁ , F ₂ , Prove that $F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right]$	7	L2	3												
18	Solve the recurrence relation a _n = na _{n-1} for n ≥ 1 given the a ₀ = 1	7	L3	3												
19	Solve the recurrence relation a _n = 7a _{n-1} for n ≥ 1 given the a ₂ = 98			3												
20	Solve the recurrence relation a _n – 3a _{n-1} = 5 × 3 ⁿ for n ≥ 1 given that a ₀ = 2	7	L3	3												
21	Determine the recurrence relation and the initial condition for the sequence, 0, 2, 6, 12, 20, 30, 42 ... Hence determine the general term of the sequence.	7	L3	3												
22	The number of virus affected files in a system is 1000(to start with) and this increases 250% every two hours. Use a recurrence relation to determine the number of virus affected files in the system after one day.	6	L3	5												
23	A bank pays a certain percent of annual interest on deposits compounding the interest once in three months. If a deposit doubles in 6 years and 6 months, determine what is the annual percent of interest paid by the bank.	6	L3	5												
24	Solve the recurrence relation a _n – 6a _{n-1} + 9a _{n-2} = 0 for n ≥ 2, given that a ₀ = 5, a ₁ = 12	7	L3	3												
25	Solve the recurrence relation a _n = 3a _{n-1} – 2a _{n-2} for n ≥ 2, given that a ₁ = 5, a ₂ = 3	7	L3	3												
26	Solve the recurrence relation a _n = 2(a _{n-1} – a _{n-2}), for n ≥ 2	7	L3	3												
27	Solve the recurrence relation F _{n+2} = F _{n+1} + F _n where n ≥ 0 and F ₀ = 0, F ₁ = 1	7	L3	3												

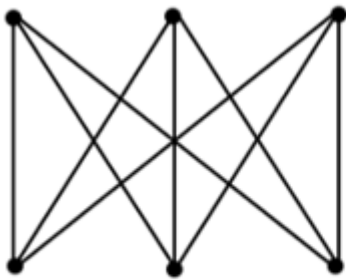
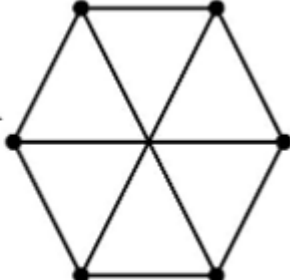
Question Bank

Course: Discrete Mathematical Structures and Graph Theory (BCSESC404A)

Module 4: Introduction to Graphs

Q. No.	Written Type Question	Marks	RL	CO
1	Define i) Regular graph ii) Trivial graph iii) Null graph with an example for each.	6	L1	CO1
2	Define i) Bipartite graph ii) Complete graph iii) Multi graph with an example for each.	6	L1	CO1
3	Define i) Parallel edges ii) Subgraph iii) Pendant vertex with an example for each.	6	L1	CO1
4	Explain any three applications of graphs.	6	L2	CO1
5	Consider the following graph G. (i) Classify the graph G based on its structural properties. (ii) Identify all pendant vertices in G. (iii) Determine the number of connected components in graph G (iv) Compute the minimum degree $\delta(G)$ of the graph G. (v) Calculate the average degree $d(G)$ of the graph G. (vi) Construct two vertex disjoint subgraphs of G.  G	7	L3	CO1
6	Calculate the number of vertices for the graph $G = (V, E)$ in the following cases: i) G has 9 edges and all the vertices of degree 3 ii) G is a cubic graph with 9 edges iii) G has 10 edges with 2 vertices of degree 4 and others of degree 3	6	L3	CO1
7	Show that the number of vertices of odd degree in a graph is always even.	7	L2	CO1
8	Show that the maximum number of edges in a simple graph with n vertices is $\frac{n(n-1)}{2}$	7	L2	CO1
9	Show that a simple graph with n vertices and k components can have at most $\frac{(n-k)(n-k+1)}{2}$ edges.	7	L2	CO1
10	Analyse the difference between Complete graph and Complete Bipartite graph.	6	L4	CO1

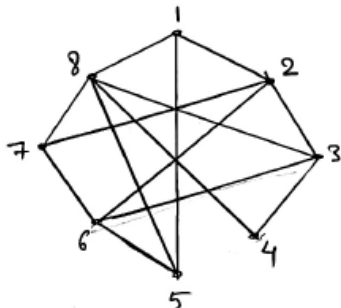
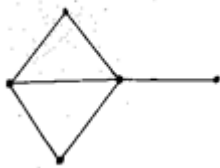
Q. No.	Written Type Question	Marks	RL	CO
11	Determine the complement of each of the following graphs  G_1  G_2  G_3	6	L3	CO1
12	Examine whether the complement of a bipartite graph need not be a bipartite graph.	6	L4	CO1
13	Analyse whether the following graphs are isomorphic or not.  G_1  G_2	7	L4	CO1
14	Analyse whether the following graphs are isomorphic or not.  G_1  G_2	7	L4	CO1
15	Analyse whether the following graphs are isomorphic or not.  G_1  G_2	7	L4	CO1

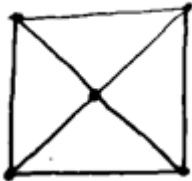
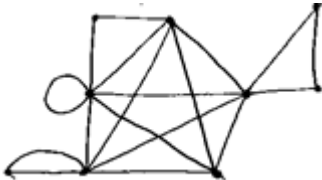
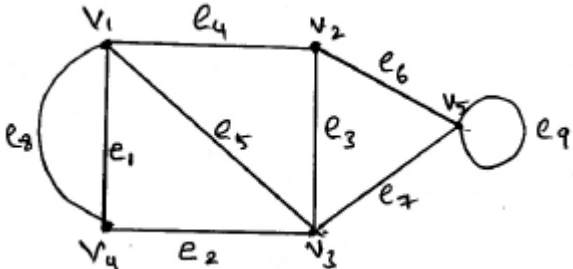
Q. No.	Written Type Question	Marks	RL	CO
16	Analyse whether the following graphs are isomorphic or not.   G_1 G_2	7	L4	CO1
17	Define i) Walk ii) Path iii) Circuit with an example for each.	6	L1	CO1
18	Define i) Unicursal graph ii) Euler graph iii) Hamilton path with an example for each.	6	L1	CO1
19	Construct two Euler graphs and an Unicursal graph by specifying the walk.	6	L3	CO4
20	Show that a connected graph G is an Euler graph if and only if all vertices of G are of even degree.	6	L2	CO4
21	Explain the Konigsberg Bridge problem in graph theory.	7	L2	CO1
22	Explain Hamilton circuit and Hamilton path with an example for each. Hence construct a graph that has a Hamilton path but not a Hamilton circuit.	7	L3	CO4
23	Explain Hamilton cycle with an example. Determine the number of edge-disjoint Hamilton cycles exist in a complete graph with 5 vertices. Hence construct the graph to show and specify the cycle.	7	L3	CO4
24	Investigate whether K_5 and Hyper cube Q_3 are Hamilton graphs by determining a Hamilton cycle in each.	6	L3	CO4
25	Construct a graph by specifying walk in each of the following cases: i) Both Euler circuit and Hamilton cycle. ii) Euler circuit but no Hamilton cycle. iii) Hamilton cycle but no Euler circuit. iv) Neither a Hamilton cycle nor an Euler circuit.	6	L3	CO4

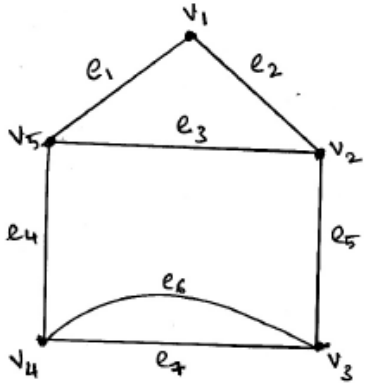
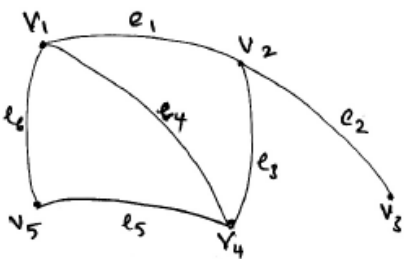
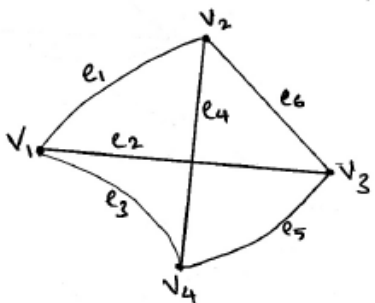
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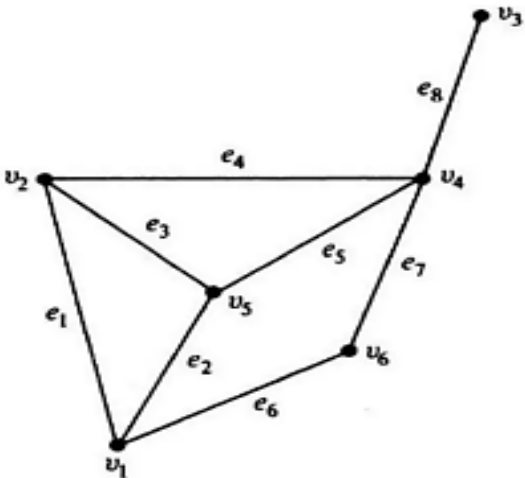
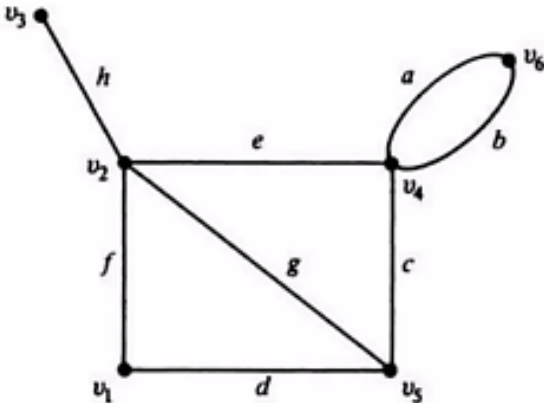
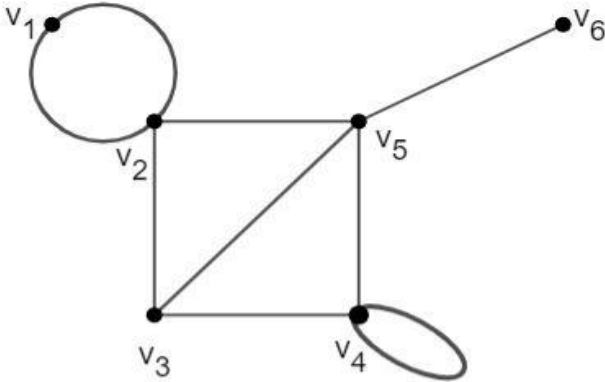
**Course Title: Discrete Mathematical Structures and Graph Theory
(BCSESC404A)**

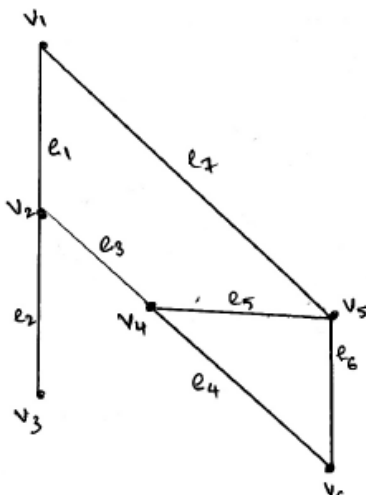
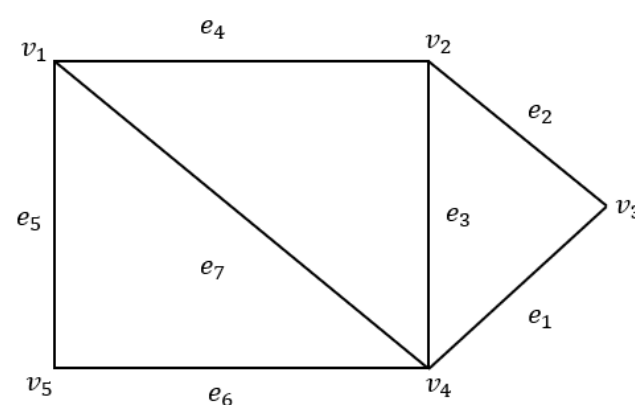
Module 5: Planar Graphs and Graph Representations

Q. No.	Written Type Question	Marks	RL	CO
1	Explain planar and non-planar graphs with an example and show that the complete graph K_4 is planar	7	L2	CO4
2	Construct a planar graph for the following cases: (a) Order 5 and size 8 (b) Order 6 and size 12	6	L3	CO4
3	Construct a graph for both planar graph and non-planar graph and give two conditions for testing planarity of any graph.	7	L3	CO4
4	Show that K_3 , $K_{2,2}$, and $K_{2,3}$ are planar graphs	6	L2	CO4
5	Explain Kuratowski's graph and draw Kuratowski's two graphs	6	L2	CO4
6	Show that Kuratowski's first graph is non planar	6	L2	CO4
7	State Kuratowski's theorem and show that Peterson graph is non planar	6	L2	CO4
8	Using Kuratowski's Theorem show that the following graph is non-planar. 	6	L2	CO4
9	Show that a connected graph G with n vertices and e edges have $e - n + 2$ regions	6	L2	CO4
10	If G is a simple planar graph with at least three vertices, then show that (i) $e \leq 3n - 6$. and (ii) $e \leq 2n - 4$; if G is triangle free.	7	L2	CO4
11	Construct the geometric dual of the following graph. 	7	L3	CO4
12	Construct the geometric dual of the following graph.	7	L3	CO4

Q. No.	Written Type Question	Marks	RL	CO
				
13	Construct the geometric dual of the following graph. 	7	L3	CO4
14	Show the planarity of the following graph by the method of elementary reduction 	7	L2	CO4
15	Show the planarity of the following graph by the method of elementary reduction? 	7	L2	CO4
16	Explain (i) Digraph (ii) Adjacency matrix (iii) Path matrix with an example for each.	6	L2	CO4
17	Show that in every digraph D, the sum of the out-degrees of all vertices is equal to the sum of the in-degrees of all vertices, each sum being equal to the number of edges in D.	6	L2	CO4
18	Explain the types digraphs with an example for each.	7	L2	CO4

Q. No.	Written Type Question	Marks	RL	CO
19	Determine the adjacency matrix and incidence matrix for the given graph. 	6	L3	CO4
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21	Determine the adjacency matrix and incidence matrix for the given graph. 	6	L3	CO4

Q. No.	Written Type Question	Marks	RL	CO
22	Determine the adjacency matrix and incidence matrix for the given graph. 	6	L3	CO4
23	Determine the Path matrix between the vertices v_2 and v_4 and Circuit matrix for the given graph. 	6	L3	CO4
24	Determine the Path matrix between the vertices v_2 and v_5 and Circuit matrix for the given graph. 	6	L3	CO4

Q. No.	Written Type Question	Marks	RL	CO
25	Determine the Path matrix between the vertices v_1 and v_5 and Circuit matrix for the given graph. 	6	L3	CO4
26	Determine the Path matrix between the vertices v_1 and v_5 and Circuit matrix for the given graph. 	6	L3	CO4

Course coordinator

Co-ordinator: DAAC

Program Coordinator